Pneumatics calculations

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the instroke force.

- Piston area = 490.9 mm², rod area = 78.5 mm²
- Effective area = 412.3 mm²
- F = 0.4 × 412.3 = 165 N

2 Write down the formula used to calculate the outstroke force of a single-acting cylinder.

- F = P × A

3 A stamping machine does not imprint its logo properly. Suggest two ways of increasing the force from its cylinder.

- Increase the air pressure
- Use a cylinder with a larger piston (larger area)

4 What controls the instroke of a single-acting cylinder?

- The return spring

5 A single-acting cylinder must press two sheets of acrylic together with a force of 300 N. The piston diameter is 20 mm and the pressure is 0.3 N/mm². Will this work? Show your working.

- $A = \pi \times 10^2 = 314$ mm²
- F = 0.3 × 314 = 94 N
- No: 94 N is less than the 300 N needed

6 What force does a 20 mm diameter cylinder produce on the outstroke at 0.8 N/mm²?

- $A = \pi \times 10^2 = 314$ mm²
- F = 0.8 × 314 = 251 N

7 A double-acting cylinder has a 60 mm piston and a 10 mm rod. The supply is 12 bar. Calculate the instroke force.

- P = 1.2 N/mm²
- $A = \pi \times 30^2 - \pi \times 5^2 = 2827 - 79 = 2749$ mm²
- F = 1.2 × 2749 = 3300 N

8 Write down the formula used to find the pressure when the force and cylinder size are known.

- P = F ÷ A

9 A stamping machine exerts 454 N with a 34 mm diameter piston. Calculate the air pressure needed, in N/mm² and in bar.

- $A = \pi \times 34^2 \div 4 = 908$ mm²
- P = 454 ÷ 908 = 0.50 N/mm² = 5 bar

10 A bottle-capping machine needs 650 N from a 56 mm diameter single-acting cylinder. Calculate the air pressure needed.

- $A = \pi \times 56^2 \div 4 = 2463$ mm²
- P = 650 ÷ 2463 = 0.26 N/mm² = 2.6 bar

11 A 60 mm diameter cylinder must push a packing case off a conveyor with a force of 540 N. Calculate the pressure needed.

- $A = \pi \times 60^2 \div 4 = 2827$ mm²
- P = 540 ÷ 2827 = 0.19 N/mm² = 1.9 bar

12 A drawer-testing rig needs 16 N from a 10 mm diameter cylinder. Calculate the pressure needed.

- $A = \pi \times 10^2 \div 4 = 78.5\ mm^2$
- $P = 16 \div 78.5 = 0.20\ N/mm^2 = 2\ bar$

13 Write down the formula used to find the piston area when the force and pressure are known.

- $A = F \div P$

14 A cylinder lifting parcels needs 180 N at 6 bar. Calculate the piston area needed.

- $P = 0.6\ N/mm^2$
- $A = 180 \div 0.6 = 300\ mm^2$

15 A sliding door needs 400 N to open. The single-acting cylinder is supplied at 5 bar. Calculate the piston diameter needed.

- $A = 400 \div 0.5 = 800\ mm^2$
- $d = \sqrt{(4 \times 800 \div \pi)} = 31.9$ mm, so a 32 mm piston

16 A furnace door weighs 100 N and is lifted by a single-acting cylinder at 4 bar. Calculate the piston diameter needed.

- $A = 100 \div 0.4 = 250\ mm^2$
- $d = \sqrt{(4 \times 250 \div \pi)} = 17.8$ mm, so an 18 mm piston